

Minimum effort and coordination incentives. Pure-strategy Nash equilibria

Consider two players with utility functions

$$u_i(x_1, x_2) = \min\{x_1, x_2\} - c_i(x_i)$$

where $x_i \in \{0, 1, 2\}$ is player i 's effort and $c_i(x_i) = x_i/2$ is its cost. Players choose their effort levels simultaneously.

- (a) Find all pure-strategy Nash equilibria. Are they efficient? Justify your answer.
- (b) Suppose the game has three players. Find all pure-strategy Nash equilibria.
- (c) Suppose the game has N players. Find all pure-strategy Nash equilibria. *Hint: use what you learned from the two- and three-player games.*
- (d) For the two-player game, explain how your answer changes if, for every player i ,

$$c_i(x_i) = \begin{cases} 0, & x_i = 0 \\ \frac{3}{2}, & x_i \in \{1, 2\} \end{cases}$$

Solution

We consider pure-strategy equilibria throughout and interpret efficiency as Pareto efficiency.

- (a) **Two players.** Each player wants to match the other's effort: choosing more does not increase the minimum of the two efforts and only raises their own cost. Choosing less reduces that cost by only half a unit per unit of effort, while reducing the minimum by a full unit, so it also lowers their payoff. The payoff matrix is shown below. Teal underlines mark player 1's best responses, and red underlines mark player 2's best responses.

$x_1 \backslash x_2$	0	1	2
0	(<u>0</u> , <u>0</u>)	$(0, -\frac{1}{2})$	$(0, -1)$
1	$(-\frac{1}{2}, 0)$	(<u>$\frac{1}{2}$</u> , <u>$\frac{1}{2}$</u>)	$(\frac{1}{2}, 0)$
2	$(-1, 0)$	$(0, \frac{1}{2})$	(<u>1</u> , <u>1</u>)

Given the opponent's effort x_j , player i 's utility is

$$u_i(x_i, x_j) = \begin{cases} \frac{x_i}{2}, & x_i \leq x_j \\ x_j - \frac{x_i}{2}, & x_i \geq x_j \end{cases}$$

Utility increases up to $x_i = x_j$ and decreases beyond it. Hence the unique best response is to match the opponent:

$$BR_i(x_j) = \{x_j\}$$

The pure-strategy Nash equilibria are

$$NE_{\text{pure}} = \{(0, 0), (1, 1), (2, 2)\}$$

Their respective payoff vectors are

$$(0, 0) \quad \left(\frac{1}{2}, \frac{1}{2}\right) = (0.5, 0.5) \quad (1, 1)$$

To assess efficiency, note that every player's payoff satisfies

$$u_i(x_1, x_2) \leq x_i - \frac{x_i}{2} = \frac{x_i}{2} \leq 1$$

At (2, 2), both players attain this upper bound. Moreover, both strictly prefer (2, 2) to either of the other equilibria.

Only (2, 2) is Pareto efficient. The equilibria (0, 0) and (1, 1) are strictly Pareto dominated by (2, 2).

- (b) **Three players.**

The natural extension of the utility function is

$$u_i(x_1, x_2, x_3) = \min\{x_1, x_2, x_3\} - \frac{x_i}{2}$$

Let $m_{-i} = \min_{j \neq i} x_j$. Then

$$u_i(x_i, x_{-i}) = \min\{x_i, m_{-i}\} - \frac{x_i}{2} \implies BR_i(x_{-i}) = \{m_{-i}\}$$

Thus, choosing the same effort is a mutual best response. Conversely, at any unequal profile, a player choosing more than the minimum can reduce effort to that minimum. This leaves output unchanged and strictly reduces their cost.

The pure-strategy Nash equilibria are exactly

$$NE_{\text{pure}} = \{(0, 0, 0), (1, 1, 1), (2, 2, 2)\}$$

(c) **N players.**

For $N \geq 2$, utilities and best responses are

$$u_i(x) = \min_{1 \leq j \leq N} x_j - \frac{x_i}{2} \quad BR_i(x_{-i}) = \left\{ \min_{j \neq i} x_j \right\}$$

If every player chooses the same $k \in \{0, 1, 2\}$, each chooses their unique best response.

To rule out unequal profiles, let $m = \min_j x_j$ and select a player i with $x_i > m$. By switching to m , this player keeps the minimum effort unchanged and obtains

$$u_i(m, x_{-i}) - u_i(x_i, x_{-i}) = \frac{x_i - m}{2} > 0$$

Therefore, an unequal profile cannot be a Nash equilibrium.

For every $N \geq 2$, there are exactly three pure-strategy Nash equilibria:

$$NE_{\text{pure}} = \{(k, \dots, k) : k \in \{0, 1, 2\}\}$$

All three are strict. Only $(2, \dots, 2)$ is Pareto efficient; it gives every player a payoff of 1 and strictly Pareto dominates the other two equilibria.

(d) **A fixed cost of positive effort.**

With the new cost function, the payoff matrix becomes

$x_1 \backslash x_2$	0	1	2
0	(<u>0</u> , <u>0</u>)	(<u>0</u> , $-\frac{3}{2}$)	(0, $-\frac{3}{2}$)
1	($-\frac{3}{2}$, <u>0</u>)	($-\frac{1}{2}$, $-\frac{1}{2}$)	($-\frac{1}{2}$, $-\frac{1}{2}$)
2	($-\frac{3}{2}$, 0)	($-\frac{1}{2}$, $-\frac{1}{2}$)	(<u>$\frac{1}{2}$</u> , <u>$\frac{1}{2}$</u>)

Against effort 0 or 1, any positive effort gives a negative payoff, whereas effort 0 gives zero. Against effort 2, choosing 2 yields $\frac{1}{2} = 0.5$, exceeding the payoff from either alternative. Hence

$$BR_i(x_j) = \begin{cases} \{0\}, & x_j \in \{0, 1\} \\ \{2\}, & x_j = 2 \end{cases}$$

The pure-strategy Nash equilibria are now

$$NE_{\text{pure}} = \{(0, 0), (2, 2)\}$$

The equilibrium $(1, 1)$ disappears: either player can improve their payoff from $-\frac{1}{2} = -0.5$ to 0 by choosing zero effort.

Under the new costs, each player's maximum possible payoff is $\frac{1}{2} = 0.5$, attained by both at $(2, 2)$.

Only $(2, 2)$ is Pareto efficient, with payoffs

$$\left(\frac{1}{2}, \frac{1}{2}\right) = (0.5, 0.5)$$

It strictly Pareto dominates the equilibrium $(0, 0)$.